Prediction of the Outcome of United State Security Clearance Appeals via NLP models

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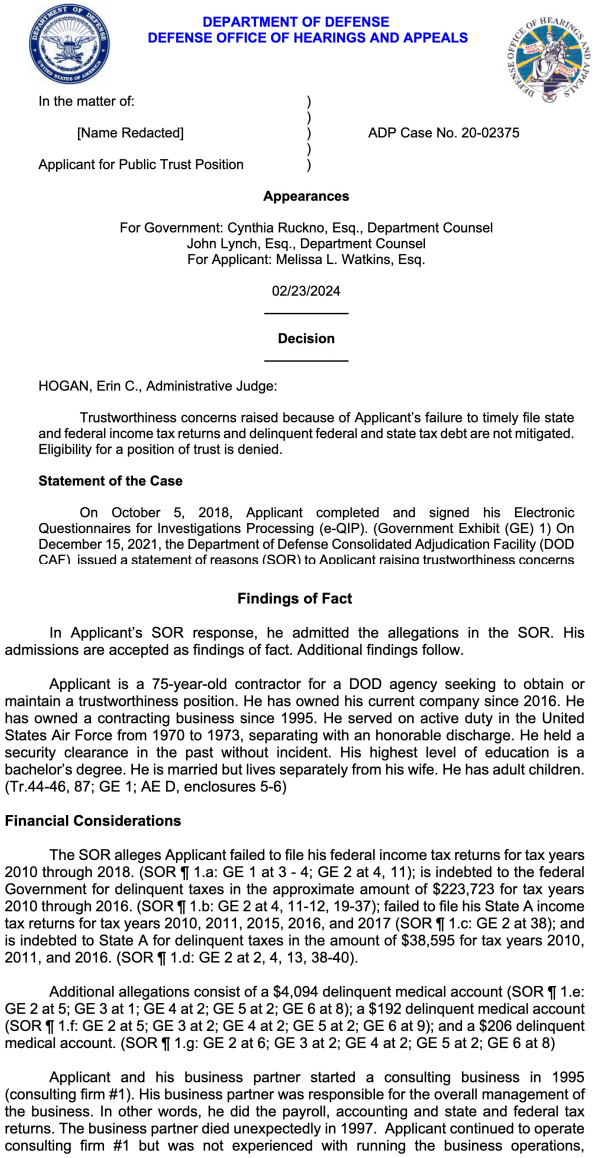
Introduction

- All persons that work with classified information in the US must get a *Security Clearance* to be able to access classified information
  - Additionally, all Federal employees require a *Public Trust* clearance
- These clearances can be denied for reasons falling under one of 13 guidelines
- If a clearance is denied, the government must issue a *Statement of Reasons* (SOR) explaining what guideline the denial references and the details leading to the denial.

(1) GUIDELINE A: Allegiance to the United States
(2) GUIDELINE B: Foreign Influence
(3) GUIDELINE C: Foreign Preference
(4) GUIDELINE D: Sexual Behavior
(5) GUIDELINE E: Personal Conduct
(6) GUIDELINE F: Financial Considerations
(7) GUIDELINE G: Alcohol Consumption
(8) GUIDELINE H: Drug Involvement and Substance Misuse
(9) GUIDELINE I: Psychological Conditions
(10) GUIDELINE J: Criminal Conduct
(11) GUIDELINE K: Handling Protected Information
(12) GUIDELINE L: Outside Activities
(13) GUIDELINE M: Use of Information Technology

Source: Security Executive Agent Directive-4, *The National Counterintelligence and Security Center*



DEPARTMENT OF DEFENSE
DEFENSE OFFICE OF HEARINGS AND APPEALS

In the matter of:	)	
	)	
[Name Redacted]	)	ADP Case No. 20-02375
	)	
Applicant for Public Trust Position	)	

Appearances

For Government: Cynthia Ruckno, Esq., Department Counsel
John Lynch, Esq., Department Counsel
For Applicant: Melissa L. Watkins, Esq.

02/23/2024

Decision

HOGAN, Erin C., Administrative Judge:

Trustworthiness concerns raised because of Applicant's failure to timely file state and federal income tax returns and delinquent federal and state tax debt are not mitigated. Eligibility for a position of trust is denied.

Statement of the Case

On October 5, 2018, Applicant completed and signed his Electronic Questionnaires for Investigations Processing (e-QIP). (Government Exhibit (GE) 1) On December 15, 2021, the Department of Defense Consolidated Adjudication Facility (DOD CAF) issued a statement of reasons (SOR) to Applicant raising trustworthiness concerns

Findings of Fact

In Applicant's SOR response, he admitted the allegations in the SOR. His admissions are accepted as findings of fact. Additional findings follow.

Applicant is a 75-year-old contractor for a DOD agency seeking to obtain or maintain a trustworthiness position. He has owned his current company since 2016. He has owned a contracting business since 1995. He served on active duty in the United States Air Force from 1970 to 1973, separating with an honorable discharge. He held a security clearance in the past without incident. His highest level of education is a bachelor's degree. He is married but lives separately from his wife. He has adult children. (Tr.44-46, 87; GE 1; AE D, enclosures 5-6)

Financial Considerations

The SOR alleges Applicant failed to file his federal income tax returns for tax years 2010 through 2018. (SOR ¶ 1.a: GE 1 at 3 - 4; GE 2 at 4, 11); is indebted to the federal Government for delinquent taxes in the approximate amount of $223,723 for tax years 2010 through 2016. (SOR ¶ 1.b: GE 2 at 4, 11-12, 19-37); failed to file his State A income tax returns for tax years 2010, 2011, 2015, 2016, and 2017 (SOR ¶ 1.c: GE 2 at 38); and is indebted to State A for delinquent taxes in the amount of $38,595 for tax years 2010, 2011, and 2016. (SOR ¶ 1.d: GE 2 at 2, 4, 13, 38-40).

Additional allegations consist of a $4,094 delinquent medical account (SOR ¶ 1.e: GE 2 at 5; GE 3 at 1; GE 4 at 2; GE 5 at 2; GE 6 at 8); a $192 delinquent medical account (SOR ¶ 1.f: GE 2 at 5; GE 3 at 2; GE 4 at 2; GE 5 at 2; GE 6 at 9); and a $206 delinquent medical account. (SOR ¶ 1.g: GE 2 at 6; GE 3 at 2; GE 4 at 2; GE 5 at 2; GE 6 at 8)

Applicant and his business partner started a consulting business in 1995 (consulting firm #1). His business partner was responsible for the overall management of the business. In other words, he did the payroll, accounting and state and federal tax returns. The business partner died unexpectedly in 1997. Applicant continued to operate consulting firm #1 but was not experienced with running the business operations,

Appealing after receiving an SOR

- Applicants have the right to appeal a denial of clearance after being issued an SOR. They will then stand trial by a judge from the *Defense Office of Hearings and Appeals* (DOHA) to make their case for why they should be granted a clearance
- These case files are posted to DOHA's website and are freely available to the public.
  - These files contain *Findings of Fact* section where the details of the clearance denial are laid out
- This project will aim to develop NLP models to predict the decision of hearings based on their *Findings of Fact*
- Two models are investigated:
  - Long Short-Term Memory (LSTM) Networks
  - Logistic Regression (LR) via TF-IDF feature extraction

Data Collection and processing

- All hearings from 2019 to the present were downloaded from DOHA's website and processed using `pdfminer.six`
- Text in parathesis was removed to take out legal citations
- For LR models only, `spacy` was used for lemmatization
- End dataset characteristics:
  - Training: 915 approvals, 2563 denials
  - Evaluation: 245 approvals, 662 denials
- Keep in mind that denial is the positive class and is represented as **1**

Handling class Imbalance

- As seen, the dataset is massively imbalanced, denials outnumber approvals 3:1
- For LR models, `sklearn` provides easy ways to apply class weights by simply passing `class_weight='balanced'`
- LSTM models class weights ended producing minimal effects, so weight sampling was used such that models would get the same proportion of deny and approve samples in a batch
  - Models can receive the same FoFs more than once in an epoch
  - May lead to overfitting, but we will see why this was not a large concern

LR Hyperparameters

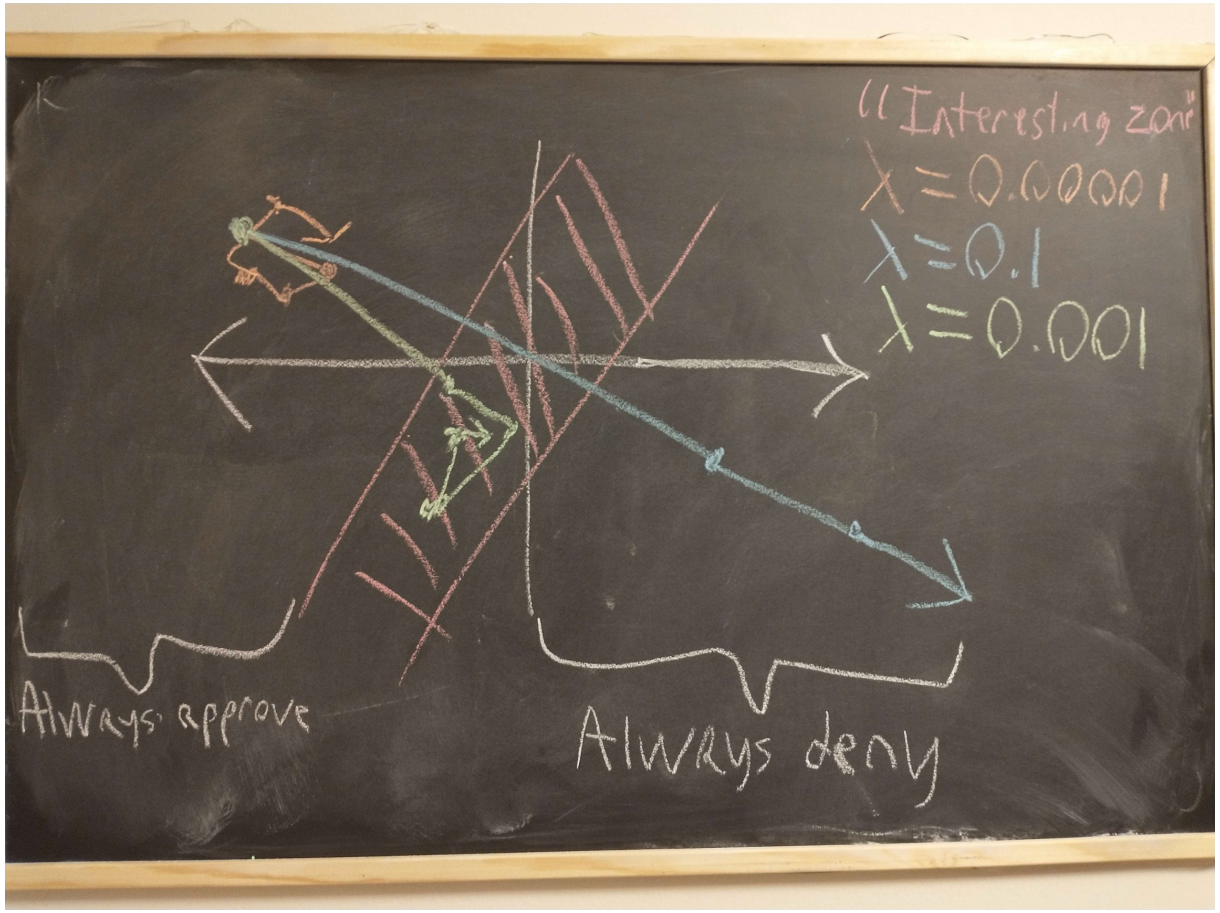
- The only LR hyperparameters modified were discrete, namely:
  - The penalty term (lasso vs elastic net vs ridge)
  - The minimum *n* for the TF-IDF vectorizer
  - The maximum *n* for the TF-IDF vectorizer
- For the LSTM models on the other hand…

Learning Rate Sensitivity for LSTM

- The “interesting” portion of weight space of the models is “narrow”
- Because of this, overfitting is not a primary concern, since the LSTM models had trouble *fitting*, let alone overfitting
    - Still tune dropout rate once a “good” learning rate is set

Example low learning rate behavior

```
P tensor([0, 0, 0, 0, 0, 0, 0, 0, 0, 0]) Batch 75 Epoch 9
T tensor([0, 0, 1, 0, 1, 0, 0, 1, 1, 1]) Batch 75 Epoch 9
A tensor(0.5000) Batch 75 Epoch 9
P tensor([0, 0, 0, 0, 1, 0, 0, 0, 1, 0]) Batch 76 Epoch 9
T tensor([0, 1, 1, 0, 1, 0, 1, 1, 0, 1]) Batch 76 Epoch 9
A tensor(0.4000) Batch 76 Epoch 9
P tensor([0, 0, 0, 0, 1, 0, 0, 0, 0, 0]) Batch 77 Epoch 9
T tensor([0, 0, 1, 1, 1, 1, 1, 1, 0, 1]) Batch 77 Epoch 9
A tensor(0.4000) Batch 77 Epoch 9
P tensor([0, 0, 0, 0, 0, 0, 0, 0, 0, 0]) Batch 78 Epoch 9
T tensor([1, 1, 0, 0, 0, 1, 0, 0, 1, 0]) Batch 78 Epoch 9
A tensor(0.6000) Batch 78 Epoch 9
```



Model Judgement

- Recall is much more important than precision
  - Don't want to give classified information to people who shouldn't have it!
  - But also, don't want to reject qualified applicants if problems are successfully mitigated
- Fβ score: modification of F1 that says recall is β times more important than precision

$$F_\beta = \frac{(1+\beta^2)\,PR}{\beta^2 P + R} = \frac{(1+\beta^2)\,t_+}{(1+\beta^2)\,t_+ + \beta^2 f_- + f_+}$$

Where

- $P = \frac{t_+}{f_+ + t_+}$ is precision
- $R = \frac{t_+}{f_- + t_+}$ is recall
- t_+, t_-, f_+, f_- are the number of true positives (correctly predicted denials), true negatives (correctly predicted approvals), false positives(incorrectly predicted denials), and false negatives (incorrectly predicted approvals)

Results

- LR outperforms all LSTM models
- Interpretation: few "red flags" determine decision
  - Can be seen by looking at n-gram weights in best LR model

Logistic Regression Results

$n_{\min}$	$n_{\max}$	`penalty`	Accuracy	Precision	Recall	F_1	F_3
1	3	Lasso	0.793	0.865	0.848	0.857	0.850
1	3	E-Net	0.723	0.863	0.737	0.795	0.748
1	3	Ridge	0.735	**0.885**	0.732	0.801	0.745
1	4	Lasso	0.793	0.863	0.851	0.857	0.853
1	4	E-Net	0.715	0.855	0.734	0.790	0.744
1	4	Ridge	0.721	0.875	0.720	0.790	0.733
2	3	Lasso	0.789	0.836	0.883	0.859	0.878
2	3	E-Net	0.693	0.841	0.714	0.772	0.725
2	3	Ridge	0.694	0.844	0.712	0.773	0.724
2	4	Lasso	**0.792**	0.828	**0.903**	**0.864**	**0.895**
2	4	E-Net	0.680	0.830	0.705	0.763	0.716
2	4	Ridge	0.674	0.829	0.697	0.757	0.709

LSTM Results

`dropout_rate`	Accuracy	Precision	Recall	F_1	F_3
0.5	0.592	**0.792**	0.598	0.681	0.613
0.1	**0.711**	0.776	**0.845**	**0.809**	**0.838**
0.001	0.658	0.782	0.738	0.759	0.742

Content Warning

- The following section will make references to in minimal detail
  - Child exploitation
  - Sexual assault
  - Sexism
  - Drug Abuse
  - Potentially Controversial Foreign Policy Issues

Largest coefficients amount $n$grams in $\mathsf{LR}(2,4,0)$

$n$gram	β_i
use marijuana	1.8993601427269533
applicant claim	1.2127381291381891
applicant state	1.1958245749981677
tax year	1.1297414776739618
applicant admit	1.0334732182654243
place collection	1.0048662271294424
answer sor	0.9559628895504945
documentary evidence	0.9437236054307672
unresolved sor	0.9425697195292493
tell investigator	0.8595430325777355
debt remain	0.8254623879692009
debt unresolved	0.7934606674303728
provide evidence	0.7887538330283138
debt sor	0.7501978263723211
government investigator	0.7320308804891255
account place collection	0.7218736670660181
marijuana use	0.7184971738696724
account place	0.7115397566459751
tax return	0.7062929143271278
make payment	0.6915360339963126

Largest negative coefficients amount $n$grams in $\mathsf{LR}(2,4,0)$

$n$gram	β_i
united states	-1.9390762444418637
applicant wife	-1.4023163656679243
united state	-1.3916602727385616
south korea	-1.1667595272041076
credibly testify	-1.1564580517093903
debt resolve	-1.0672568886122413
monthly payment	-1.050131772431794
student loan	-0.8533767686892256
citizen resident	-0.8482770813321437
applicant credibly	-0.8213614831050416
evidence documentation	-0.7674057070988334
applicant contact	-0.7636224962466511
documentation creditor	-0.7345304843700309
applicant husband	-0.6628623776244902
state applicant	-0.6050158721672176
immigrate united	-0.5792400164302883
debt resolve sor	-0.5664899670881716
applicant work	-0.5582412875564088
applicant pay	-0.5552669260689507

Problems of Model

- Gender (and other sociological) bias
  - "applicant wife" has greater effect on approval than "applicant husband"
- Underestimation of how problematic some ngrams are due to rarity
  - "use cocaine" and "use lsd" have coefficients 100% less in magnitude than "use marijuana"
  - "child pornography" has a lesser magnitude coefficient than all the above drug ngrams
  - "sexual harassment" has a *negative* coefficient, meaning it *increases* the chance of clearance approval

Quick digression on Foreign Influence

Coefficients of "{country_adj} citizen" in LR$(2, 4, 0)$

Friendly Country	β_i	Hostile Country	β_i
Ukraine	-0.0183	Russia	0.0104
Taiwan	-0.04091	China	0.0223
Colombia	0.0183	Venezuela	0.0075
Israel	0.0135	Iran	-0.06230

Conclusion

- LSTM models are outperformed by LR models due to concentration of denials in a few “red flags”
- Best LR preform well, achieving recall of 90% and F3 of 0.89
  - But it is clear the model is far from robust

Suggestions for improvement

- Utilize transformer models that can be fed clearance guidelines before fitting
- Adopt “Rationale Extraction” paradigm common in other legal judgement models
- Divide model into sub-models for each security guideline
- Clean data of information on protected characteristics (except nationality, for obvious reasons)